MILITARY SPECIFICATION

CAP, EXTREME COLD WEATHER: WITH EXTENSION FLAP

This specification is approved for use by U. S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a fully lined and insulated cold weather head covering with extension flap, forehead flap and earlaps.

1.2 Classification. The caps shall be of one type in the following sizes (see 6.2):

Schedule of sizes

6-1/2	7-1/4
6-3/4	7-1/2
7	7-3/4

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps, Headquarters, U. S. Marine Corps (LMA-2), Washington, D. C. 20380, by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8415

AMSC N/A

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SPECIFICATIONS

FEDERAL

V-T-285	- Thread, Polyester
NN-P-71	- Pallet, Materials Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
UU-P-31	- Paper, General Specifications and Methods of Testing
CCC-C-458	- Cloth, Flannel, Cotton
DDD-L-20	- Label: For Clothing, Equipage, and Tentage (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-M-5631	- Mouton (for Flight Clothing)
MIL-P-15011	- Pallet, Material Handling, Wood, Post Construction, 4 Way Entry
MIL-C-17157	- Cloth, Knitted, Cotton (Waffle Type)
MIL-F-21840	- Fastener Tape, Hook and Pile, Synthetic
MIL-C-43468	- Cloth, Camouflage Pattern, Wind Resistant Poplin, Cotton
MIL-C-44031	- Cloth, Twill, Camouflage Pattern, Cotton and Nylon

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

LAWS AND REGULATIONS

U. S. Postal Service Manual

(Application for copies of the manual may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C 20402.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951 STANDARD PRACTICE FOR COMMERCIAL PACKAGING

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard samples. Standard samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.4.)

3.3 Materials

3.3.1 Basic material. The basic material for the crown, crown lining, earlap, extension flap, forehead flap, visor and binding shall be woodland camouflage pattern cotton/nylon twill cloth conforming to class 1 of MIL-C-44031 or woodland camouflage pattern cotton ripstop poplin cloth conforming to type III of MIL-C-43468. (see 6.2)

3.3.2 Lining for crown lining, forehead flap and extension flap. The lining material shall be cotton flannel cloth, Olive Green 107, conforming to type III, class 3 of CCC-C-458.

3.3.3 Lining for earlap. The material for lining the earlap shall be mouton, type II, conforming to MIL-M-5631 and shall be dyed Olive Green 107. The mouton shall be 5/8 + 1/16 inch in height.

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3.3.4 Insulation for crown. The material for insulating the crown shall be waffle type cotton knit cloth conforming to MIL-C-17157.

3.3.5 Earlap fasteners. The material for the adjustable closure shall be nylon fastener tape, 1 inch wide pile tape and 1 inch wide hook tape, conforming to type II, class 1 of MIL F-21840. The color shall be Olive Green 106.

3.3.6 Visor interlining. The visor interlining shall be made out of a nonwoven material produced from a highly purified alpha cellulose vegetable fiber, wet web impregnated with approximately 22 + 5 percent compounded neoprene and shall conform to the following physical requirements, when tested as specified in section 4, Table III.

Requirement	Machine Direction	Crosswise Direction
Stiffness (load lb. min)		
Original	2.4	1.5
After laundering	2.4	1.5
After wet (dry) cleaning	2.2	1.4
Tensile strength (lb per in, min)	156	103
Stretch (percent max)	11	22
Tearing strength (gms min)	1770	2040
Weight (lb per sq yd) 2.21 ± 5 percent		
Thickness 0.075 ± 0.003 inch		

3.3.6.1 Water absorption. The material shall not absorb more than 30 percent water when tested as specified in section 4, Table III.

3.3.6.2 Resistance to perspiration. The material shall show no embrittlement, development of stickiness, or delamination when subjected to the perspiration test specified in section 4, Table III.

3.3.6.3 Accelerated aging. The material shall show no embrittlement, development of stickiness, or delamination after 15 days in an oven with circulating air at 100°C when tested as specified in section 4, Table III.

3.3.6.4 Resistance to water. The material shall not shrink or warp after soaking in water 24 hours when tested as specified in section 4, Table III.

3.3.6.5 Directional marking. The finished alpha cellulose material shall be marked by directional symbols to designate the greater flexibility of the material. The marking shall be legibly and clearly designated on the textured side of the material.

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3.3.7 Thread. The thread for all seaming and stitching shall be polyester, size B, 2 or 3 ply Olive drab shade S-1, Cable No. 66022 conforming to type I, class 1 of V-T-285.

3.3.7.1 Colorfastness. The dyed thread shall show fastness to light and laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to light and laundering.

3.3.8 Label. Each cap shall have a combination size, identification, and instruction label conforming to type VI, class 14 of DDD-L-20. The printing shall show fastness to laundering and shall bear the following inscription:

Cap, Extreme Cold Weather with Extension Flap
Contract No: DLA-100-0-000 (Example)
Stock No: 8415-00-000-0000 (Example)
Size: 6-1/2 (Example)
Contractor:

Cleaning Instructions

1. Wash cap in warm water (80 degrees to 100 degrees Fahrenheit) using mild detergent.

2. Rinse thoroughly 2 or 3 times in cold water. Air dry.

3. -or- DRY CLEAN

3.4 Design. The cap shall be fully a lined, cold weather head covering having a turndown forehead flap (sinus protector), six piece outer crown, extension flap and earlaps with nylon fastener hook and pile tape overlap closure (see Figures 1 through 5).

3.5 Patterns. Standard patterns providing a 1/4-inch seam allowance for all seams (unless otherwise specified) will be furnished by the Government. The Government patterns, which show directional lines and markings for proper assembly, shall not be altered in any way and are to be used as a guide for cutting supplier's working patterns. The working patterns shall be identical to the Government patterns, except that additional notching to facilitate manufacture is permissible.

3.5.1 Pattern parts. The components of the cap shall be cut from the materials, as specified below, in accordance with the pattern parts indicated.

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Item	Pattern Nomenclature	Number of Cut Parts
	Material: Basic Material	
1.	Top Crown - Front	2
2.	Top Crown - Side	2
3.	Top Crown - Back	2
4.	Earlap	2
5.	Forehead Flap	1
6.	Extension Flap (Ear)	1
7.	Visor	2
8.	Inner Crown - Side	2
9.	Inner Crown - Top	2
	Material: Lining, Cotton Flannel	
10.	Inner Crown - Side	2
11.	Inner Crown - Top	2
12.	Forehead flap	1
13.	Extension flap (Ear)	1
	Material: Insulating, Cotton Knit, Waffle Type	
14.	Crown	1
	Material: Lining (Fur)	
15.	Earlap	2
	Material: Interlining	
16.	Visor	1
	Material: Hook and Pile	
17.	Earlap pile and hook	1

3.6 Construction. The construction shall conform in all respects to the requirements specified in Table I and herein. Figure 1 through 5 are furnished solely for guidance and information. Should variation from the document appear in the figures, the document shall govern. The cap shall be manufactured in accordance with operations requirements specified in Table I. The contractor is not required to follow the exact sequence of operations listed.

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3.6.1 Stitches, seams, and stitchings. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in Table I. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, either one may be used. Any additional basting or holding stitching to facilitate manufacture is permissible provided the thread is removed or does not show on the finished cap.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/4 inch except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.1.1 Repairs of type 301 stitching.

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during sewing, except prestitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch in back of the end of the stitching. 1/

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

3.6.2 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.6.3 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the cap.

3.6.4 Shade marking. The component parts of the cap, except the sweatband and underply of curtain shall be marked to insure a uniform shade and size throughout the cap. Any method of shade marking may be used except:

1. Corrosive metal fastening devices.
2. Sew-on shade tickets.
3. Adhesive type tickets whose adhesive discolors or adheres to the material after removal of the tickets.

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

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TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
1.	Cutting. a. Cut the material in strict accordance with patterns furnished which show size, shape, warp or filling directional lines, and notches for proper assembly. The crown pieces and the crown lining pieces may be cut in either the warp or filling direction. b. Cut the basic material parts from the same piece of material except the forehead flap, inner crown, extension flap and binding which may be cut from ends. Cut the lining for the earlaps from the same piece of material. c. Cut binding for visor from the basic material, on the bias 1 inch wide. d. Cut the forehead flap lining, extension flap lining and inner crown lining (flannel) in either the warp or filling direction. e. Cut the visor interlinings from the nonwoven cloth in accordance with the pattern. f. Cut the cotton knit interlining for the crown in accordance with the pattern. g. Cut the nylon fastener tapes by the pattern.					
2.	Replacement of damaged parts. a. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.2.4.1 shall be removed from production and replaced with nondefective and properly matched components.					

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TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
3.	Component marking.					
	a. Mark, ticket or bundle all parts, to insure a uniform shade, size and proper assembly throughout the cap.					
4.	Make extension flap.					
	Finished appearance. The extension flap shall be neatly joined without twists, puckers or pleats. The flannel lining shall not be rolled outside of basic fabric.					
	a. Stitch the darts in the extension flap as indicated by marks on the pattern.	301	OSF-1	8-12	50-2	50-2
	b. Stitch darts in the extension flap lining as indicated by marks on the pattern.	301	OSF-1	8-12	50-2	50-2
	c. Position the extension flap and flap lining, face to face, and stitch together with a 1/4-inch seam; turn, forcing out the edges, and raise stitch 1/8 inch from turned edges.	301	SSe-2	10-14	50-2	50-2
	d. Join the top raw edges of the extension flap with a 1/4-inch seam	301	SSa-1	8-12	50-2	50-2
5.	Make earlaps and earlap lining.					
	Finished appearance. The earlaps shall be neatly joined without twists, puckers, pleats or excessive fullness.					

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TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
5.	Make earlaps and earlap lining. (cont'd)					
	a. Stitch the darts on the earlaps as indicated by marks on the pattern, extending the stitching 1/2 inch beyond end of dart.	301	OSF-1	8-12	50-2	50-2
	b. Position the hook fastener tape to the right earlap in accordance with the marks on the pattern. Stitch all edges 1/16 inch from the edge of tape, over-lapping the stitching ends not less than 1 inch.					
	c. Position the earlap pieces face to face and join the back seam with a 1/4-inch seam.	301	SSa-1	10-14	50-2	50-2
	d. Position the pile fastener tape to the left earlap lining in accordance with the marks on the pattern. Stitch all edges 1/16 inch from the edge of fastener tape, overlapping the stitching ends not less than 1 inch.	301	LSbj-1	10-14	50-2	50-2
	e. Position the earlap lining pieces face to face and join the back seam with a 1/4-inch seam.	301	SSa-1	10-14	50-2	50-2
	f. Position the earlap shell and earlap lining face to face and stitch raw edges together 1/4 inch from edge along sides and across bottom. Turn, forcing out the edges and edge stitch with a 1/4-inch seam.	301	SSe-2	10-14	50-2	50-2
	g. The top raw edges of the earlap shall be seamed together with a 1/8-inch seam.	301	SSa-1	8-12	50-2	50-2
6.	Make forehead flap.					
	Finished appearance. The forehead flap shall be smooth and free of wrinkles and puckers. Raised stitching shall be neat and evenly gaged.					

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TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
6.	Make forehead flap. (con't)					
	a. Position the forehead flap and flap lining, face to face, and stitch the outer plies together with a 1/4 inch seam along the curved edge; turn, forcing out the edges, and raise stitch 1/8 inch from edge.	301	SSe-2	10-14	50-2	50-2
	b. The top raw edges shall be seamed together with a 1/8-inch seam. The edges shall be even.	301	SSa-1	8-12	50-2	50-2
7.	Make visor.					
	Finished appearance. The visor is constructed of two plies of basic material, one ply of non-woven cloth and a binding of basic fabric.					
	a. Position the nonwoven cloth between the two plies of basic material and stitch 1/8 inch from outer edges of visor.	301	SSa-1	8-12	50-2	50-2
	b. Quilt visor with 6 rows of stitching through all plies of material, 1/4 inch apart, with the design properly centered on the visor. The first row of stitching shall be 1/2 inch from the outer raw edge of visor.	301	SSv-6	10-12	50-2	50-2
	c. Bind outer edge of visor. The binding shall have folded edges and finish 1/4 inch in width.	301	BSc-1	10-12	50-2	50-2
	NOTE: No splicing shall be permitted on the visor binding.					
	d. Stitch inner raw edge of visor with a 1/8 inch seam.	301	SSa-1	8-12	50-2	50-2
	e. Notch raw edge at center of visor.					

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TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
8.	Make outer crown.					
	Finished appearance. The outer crown shall be neatly joined without twists, puckers or pleats and with seams matching at peak of crown. Raised stitching shall be neatly and evenly gaged. The crown shall be well shaped.					
	a. Position three outer crown sections together to make one half of crown assembly.					
	b. Join each section together, matching the notches, with a 1/4-inch seam; turn seam toward back and raise stitch 1/16 inch from folded edge.	301	LSq-2	10-14	50-2	50-2
	c. To make the outer crown, join two side assemblies together with a 1/4-inch seam; turn seam and raise stitch 1/16 inch from folded edge.	301	LSq-2	10-14	50-2	50-2
	NOTE: The joining seam of the two side assemblies shall be the center seam of the outer crown.					
9.	Make inner crown with lining.					
	Finished appearance. The inner crown is constructed of one ply of basic material and one ply of cotton flannel material. The lining pieces shall be neatly joined without twists, puckers, or pleats and with center joining seam matching the outer crown center joining seam. The top and side pieces shall be well shaped. The label shall be centered, and the stitching shall not be through the printing.					

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TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
9.	Make inner crown with lining. (cont'd)					
	a. Position the flannel lining panels on the respective inner crown panels.					
	b. Join a side panel to a top panel, matching the notches, with a 1/4-inch seam; turn the seam toward top of crown and raise stitch 1/8 inch from seamed edges. This will be one side assembly.	301	LSq-2	10-14	50-2	50-2
	c. Join two side assemblies, matching notches, with a 1/4-inch seam; turn and raise stitch 1/8 inch from folded edge.	301	LSq-2	10-14	50-2	50-2
	NOTE: The joining seam of the two side assemblies shall be the center seam of the inner crown.					
	d. Center the label on the back center joining seam, 1-1/2 inches from outer raw edges, and stitch all four sides.	301	LSbj-1	10-14	50-2	50-2
10.	Make insulated crown and attach to inner crown.					
	a. Abut the edges of the cut out dart and stitch with a 1/4-inch seam.	301	SSa-1	8-12	50-2	50-2
	b. Position the insulated crown on the inner crown, matching darts and side joining seams.					
	c. Join the insulated crown to the crown lining 1-7/8 inches from bottom raw edge of inner crown.	301	SSa-1	10-14	50-2	50-2
	NOTE: The insulated crown shall fit smoothly over the inner crown lining.					

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TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
11.	Attach visor and earlaps to outer crown.					
	a. Position the raw edges of the visor and earlap to the raw edge of the outer crown assembly, matching the notches and join with a 1/8-inch seam.	301	SSa-1	10-14	50-2	50-2
12.	Attach extension and forehead flaps to inner crown.					
	a. Position the raw edges of the extension and forehead flaps to the raw edges of the inner crown assembly, matching the notches, and join with a 1/8-inch seam.	301	SSa-1	10-14	50-2	50-2
13.	Assemble cap.					
	Finished appearance. The outer crown and inner crown assemblies shall be neatly joined without twists, puckers or pleats. The crown shall have a smooth appearance.					
	a. Position the outer crown assembly and the inner crown assembly face to face, matching the notches, center seams and raw edges. Join the assemblies together with a 1/4-inch seam, leaving an opening the length of the forehead flap between the notches.	301	SSa-1	10-14	50-2	50-2
	b. Turn the crown assembly right side out.					
	c. Finish the opening by folding the visor over the crown, with the earlaps, forehead and extension flaps laying flat. Insure that the visor and forehead flap raw edges are inside the crown plies 1/8 inch from folded edge of visor and that forehead flap begins and ends 1/2 inch beyond the opening.	301	SSa-1	10-14	50-2	50-2

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TABLE I MANUFACTURING REQUIREMENTS

NO	OPERATION	STCH TYPE	SEAM/ STCH TYPE	STCH/ IN	NDL	BOB/ LPR
13.	Assemble cap. (cont'd)					
	d. Sew a stand through the visor and crown plies only, starting at end of visor on finishing seam line, increasing stitch depth to 1/2 inch at center of visor and reducing to stitch line at opposite end of visor.	301	SSa-1	10-14	50-2	50-2
14.	Clean cap.					
	a. Trim all thread ends to 1/4 inch maximum and remove loose threads on inside and outside of cap.					
	b. Remove all shade markings from the cap.					
	c. Clean all spots and stains.					

3.7 Finished measurements. The finished caps shall conform to measurements shown in Table II.

TABLE II. Cap measurements in inches 1/

	Cap size			Tolerance		
	6-1/2	6-3/4	7	7-1/4	7-1/2	7-3/4
Inside circumference 2/	21	21-3/4	22-1/2	23-1/4	24	24-3/4

1/ Tolerance ± 1/4 inch

2/ Measure the inside circumference of crown at the closing seam line.

3.8 Workmanship. The finished cap shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL's).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).

2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in MIL-STD-105.

4.4 First article inspection. When first article is required, inspection shall be made of a completely fabricated cap for all provisions of this specification.

4.5 Quality conformance inspection.

4.5.1 Testing of components. Testing of components shall be in accordance with the applicable subsidiary specifications and standards to the extent applicable, except that this specification shall govern in the event of conflict. The method of testing shall be as specified in FED-STD-191, wherever applicable. Testing shall be performed on components listed in Table III for characteristics noted.

4.5.2 In-process inspection. Inspection shall be performed during the manufacturing of the end item to ascertain that the operations are performed as specified herein. The Government reserves the right to determine the acceptability of construction methods not performed in accordance with specified requirements for operations or subassemblies.

4.5.3 Sampling for testing. Unless otherwise specified in subsidiary specifications, sampling shall be in accordance with the following:

Lot Size (lot unit)	Sample size (sample unit)
800 or less	2
801 to 22,000	3
22,001 and over	5

The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified herein. The unit for expressing lot size and sample unit for testing component shall be in accordance with applicable subsidiary specification and as follows:

Component	Lot size expressed as	Sample unit for testing
Visor interlining	Yard	1 square yard

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Table III. Test methods

Components	Characteristics	Rqmt. para.	Test method	No. Determinations per sample unit	Results reported as:
Visor interlining	Material identification	3.3.6	1/	1	Pass or fail
	Compounded neoprene content	3.3.6	1/	1	To the nearest Percent
	Stiffness:				
	Original	3.3.6	5202 2/		
	After laundering	3.3.6	5610/5202 2/		
	After wet (dry) cleaning	3.3.6	5622/5202 2/		
	Tensile strength	3.3.6	171 3/	10	To the nearest
	Stretch	3.3.6	171 3/	10	To the nearest
	Tearing Strength	3.3.6	170 3/	5	To the nearest
	Weight	3.3.6	5041 2/	1	To the nearest
	Thickness	3.3.6	5030 2/	1	To the nearest
	Water absorption	3.3.6.1	5502 2/	1	Pass or fail
	Perspiration	3.3.6.2	5680 2/	1	Pass or fail
	Accelerated aging	3.3.6.3	4/	1	Pass or fail
	Shrinkage	3.3.6.4	5/	1	Pass or fail

1/ A certificate of compliance is required and will be accepted for the requirement stated. (see 4.1.1)

2/ FED-STD-191.

3/ UU-P-31.

4/ The material shall be kept in a forced circulating air over 15 days at 100°C.

5/ The material shall be soaked in water at room temperature for 24 hours.

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4.5.4 Examination of the end item. The end item shalll be examined for defects listed in 4.5.4.1 and 4.5.4.2. The applicable inspection levels and AQL's shall be as indicated in 4.5.4.4. The lot size shall be expressed in units of caps. The sample unit shall be one finished cap. All defects shall be classified as indicated, except those with asterisk (*) in the classification column shall be classified as follows:

Major defect - When seriously affecting appearance or serviceability.

Minor defect - When affecting appearance or serviceability but not seriously.

4.5.4.1 General defects. General defects shall be classified as follows:

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
Material defects and damages	a. Hole, cut, tear, or needle chew:			
	1. On outside.	X		
	2. On inside:			
	- up to 1/8 inch in diameter or length.		X	
	- more than 1/8 inch.	X		
	b. Mend, darn, or patch:			
	1. On outside:			
	- any.	X		
	2. On inside:			
	- up to 1/2 inch in diameter or length.			X
	- more than 1/2 inch.		X	
	c. Friction burn:			
	- when fibers are broken, use hole, cut, tear or needle chew classification.			
	- more than 1/4 inch in length.			X
	d. Weakening defects:			
	1. Smash.	X		
	2. Multiple float or loose slub:			
	- on outside.		X	
	- on inside.			X
	e. Unsightly slub or shade bar:			
	- on outside.		X	
	- on inside.			X
	f. Drill hole exposed on outside.	X		
Shaded parts	a. Outside part shaded.		X	
Cleanness	a. Any spot or stain on outside.			X
	b. One or more shade tickets not removed.			X

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EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
	c. Any shade marking visible on outside.			X
	d. Thread ends not trimmed, or loose threads not removed throughout cap.			X
Component and assembly	Any component part or required operation omitted (unless otherwise classified herein).	X		
Cutting	Any component part not cut in conformance with directional lines indicated on patterns or not in accordance with specification requirements.	X		
Seams and stitching	a. Accuracy of seaming 1. Seam twisted, puckered or pleated.		X	
	2. Part of material caught in unrelated operation or stitching.	X		
	3. Stitching overlapping end(s) of thread break less than 1/2 inch.			X
	4. Ends of stitching not secured as specified.		X	
	b. Gage of stitching: 1. Irregular, i.e., unevenly gaged or corresponding stitching not uniform (to be scored only when condition exists along major portion of seam).		X	
	2. Not within range of gage specified or varies more than 1/16 inch when no range is specified.			X
	c. Stitches broken or missing: - up to 1/4 inch inclusive.			X
	- more than 1/4 inch but not more than 1/2 inch.		X	
	- more than 1/2 inch.	X		
	d. Skipped stitches: - 1/4 inch to 1/2 inch long		X	
	- more than 1/2 inch long.	X		

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EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
	e. Runoffs: - 1/4 inch to 1/2 inch long. - more than 1/2 inch long.		 X	X
	f. Raw edge: - more than 1/8 inch long.			X
	g. Seam and stitch type: - seam type or stitch type not as specified.			X
	h. Stitches per inch: - more than maximum or less than minimum specified.			X
	i. Stitch tension: 1. Loose tension resulting in a loose seam. 2. Loose tension (exposed loops of top or bottom threads): - on outside. - on inside. 3. Tight tension (stitches break when strain is applied in the direction of the seam or stitching.)	X	 X X	 X
	j. Open seams: - up to and including 1/4 inch. - more than 1/4 inch.	 X	X	
Labels	a. Missing, incorrect, or illegible: - combination identification-size-instruction label.	X		
	b. Label not stitched on all four sides.			X
	c. Label not positioned as specified.			X
	d. Stitching through the printing on label.			X
	e. Ends of stitching overlapped less than three stitches.			X

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4.5.4.2 Detailed defects. Detailed defects shall be classified as follows:

EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
Outside crown Forehead flap	a. Fleece side of forehead flap on the underside, when flap is in the "up" position.	X		
	b. One or both ends of forehead flap not overlapped with crown-to-earlap joining seam by more than 1/8 inch.			X
	c. Forehead flap not stitched to cap through all plies.	X		
	d. Forehead flap not centered over face opening by more than 1/4 inch.		X	
Visor	a. Binding twisted or stretched (wavy) on visor.		X	
	b. Quilting stitches not properly shaped.		X	
	c. Binding not securely caught in fabric (binding can be pulled from fabric).	X		
	d. Binding less than 3/16 or more than 5/16 inch wide.			X
	e. Upper or lower plies of material badly wrinkled or puckered, pleated or excessively full between rows of quilting stitches.	X		
	f. Number of rows of quilting stitches (including row at finished edge of visor):			
	- 5 or less.	X		
	- 7 or more.		X	
	g. Visor not centered with center front of cap:			
	- more than 1/4 inch.	X		

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EXAMINE	DEFECT	CLASSIFICATION		
		MAJOR	MINOR	
			A	B
	h. Edge of visor poorly shaped, affecting appearance.		X	
	i. Width of visor at center less than 2-1/2 inches or more than 2-3/4 inches.			X
	j. Binding spliced.			X
Earlap	a. Fastener tape(s) not securely sewn.	X		
	b. End(s) of stitching on fastener tape(s) overlapped less than 1 inch.		X	
	c. Pile lining of earlap on the underside when earlap is in the "up" position.	X		
	d. Pile portion of fastener tape not on inside of left earlap.	X		
	e. Hook portion of fastener tape not on outside of right earlap.	X		
	f. Fastener tape(s) not properly positioned.		X	
Crown	a. Outside ply(s) tight or full.	X		
	b. Seams do not match at peak - up to 1/4 inch inclusive. - more than 1/4 inch but not more than 1/2 inch. - 1/2 inch or more.	X	X	X
Inside cap	a. Insulation of crown lining omitted.	X		
	b. Lining(s) (crown and earlap) tight or full.		X	
	c. Extension flaps twisted or puckered.	X		
	d. Extension flap attached with flannel lining outside.	X		
	e. Flannel lining omitted.	X		
	f. Flannel lining not smooth.		X	
	g. Center seam not centered on outer crown and visor.		X	

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4.5.4.3 Dimensional examination. The end item shall be examined for conformance to the dimensions specified in Table II. Any dimensions deviating from the specified requirements shall be classified as defects.

4.5.4.4 Inspection levels and AQL's. The inspection levels and the AQL's expressed in defects per one hundred units (DHU) for visual and dimensional examination shall be as follows:

4.5.4.1 and 4.5.4.2

Inspection Level III of MIL-STD-105

AQL Major	1.5 DHU
AQL Major and Minor A (combined)	10.0 DHU
Total Major Minor A and Minor B (combined)	15.0 DHU

4.5.4.3

Inspection Level S-3 of MIL-STD-105

One Class	4.0 DHU

4.5.5 Examination of packaging requirements. An inspection shall be made to determine that preservation, packing and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully packaged. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of DHU's shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Bulged or distorted container. Bundle not securely tied in the manner specified.
Contents	Number of bundles per container is more or less than required. Number of caps per bundle is more or less than required. 1/ Size shown on one or more caps not as specified on outer container. 1/

1/ For this defect, two bundles from each container in the sample shall be examined.

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4.5.6 Examination of palletization. An examination shall be made to determine that the palletization complies with Section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully prepared for delivery. The lot size shall be the number of palletized unit loads in the end item inspection lot. The inspection level shall be S-1 and the AQL shall be 6.5 DHU's.

Examine	Defect
Finished dimension	Length, width or height exceeds specified maximum requirement.
Palletization	Not as specified.
	Container pattern not as specified.
	Interlocking of loads not as specified.
	Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each cap shall be laid flat on its side with the unfastened earlap extensions superimposed. The earlap extensions shall be folded together back over on the earlap. Five folded caps of one size only shall be neatly stacked in a bundle, with every other one reversed (bottom to crown), and securely crosstied with cotton tape or twine.

5.1.2 Commercial. The caps shall be packaged in accordance with ASTM-D-3951-82.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

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5.2.1 Level A. Eighty caps of one size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each ship- ping container shall be fitted with a box liner conforming to type CFR, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A bundles shall be packed flat two in length, two in width, and four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. In addition, each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper. Each shipping container shall be closed in accordance with Method III, waterproofed in accordance with Method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.5. Towards the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed in the same container.

5.2.2 Level B.

5.2.2.1 Method 1. Eighty caps of one class and size only, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to Style RSC-L, Type CF (variety SW) or SF, Class Domestic, Grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to Type CF, Class Domestic, variety DW, Grade 275 of PPP-B-636. Level A bundles shall be packed flat two in length, two in width and four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. In addition, each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper. Each shipping container shall be closed in accordance with Method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.5.5. Towards the end of the contract, or when there are less than the required amount per container of the same size, mixed sizes may be packed in the same container.

5.2.2.2 Method 2. When specified (see 6.2), the shipping container shall be a Grade V3c, V3s, or V4s, fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with Method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.5.5.

5.2.3 Commercial packing. The caps, packaged as specified in 5.1, shall be packed in accordance with ASTM-D-3951-82.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM-D-3951-82, as applicable.

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5.3.1 Mixed sizes. Each shipping container, packed with mixed sizes only, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches, with the words "Mixed Sizes" plainly printed or stamped thereon, and under these words shall be legibly stamped or printed the correct quantity and sizes contained therein.

5.4 Palletization. When specified (see 6.2), caps packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type 1A of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C, K and L or O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, Type I, Class 1, Style 1, Size A, Wood Group I, II, III, or IV of MIL-P-15011, or, 4-way, Type IV, V or VIII, Class 1, Style A, Size 2, Wood Group I, II, III, or IV, Grade A of NN-P-71. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall be approved by the contracting officer.

6. NOTES

6.1 Intended use. The cap is intended for wear by military personnel of the U.S. Marine Corps and the Department of Defense.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this document.
(b) Size required (see 1.2).
(c) When first articles are not required (see 3.2)
(d) Type of material required.
(e) Selection of applicable levels of preservation and packaging (see 5.1 and 5.2).
(f) When level B, Method 2 packing is required (see 5.2.2.2).
(g) When palletization is required (see 5.4).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

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Paragraph No.	Data requirements	Applicable DID No.	Options
(a) 4	Inspection system program plan	DI-R-4803	___
(b) 4.4	First Article Inspection Procedures	DI-T-4901	___
(c) 4.4	First Article Inspection Report	DI-T-4902	___
(d) 4.5	Production/Acceptance Inspection Procedures	DI-T-4903	___
(e) 4.5	Production Inspection Report	DI-T-4904	___

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000., 19L., Volume II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 Samples. For access to samples, address the acquiring activity issuing the invitation for bids.

6.4 First article. Examination, tests and approval shall be as specified by the contracting officer (see 3.2).

6.5. Subject term (key word) listing.

Camouflage

Cap

Cold weather

Earlaps

6.6 Figures. Figures 1 through 5 show the general design of the cap and are furnished for information only. In the event of conflict between the figures and the text of this specification, the text of this specification shall govern.

Preparing activity
Navy-MC
Project No. 8415-N549

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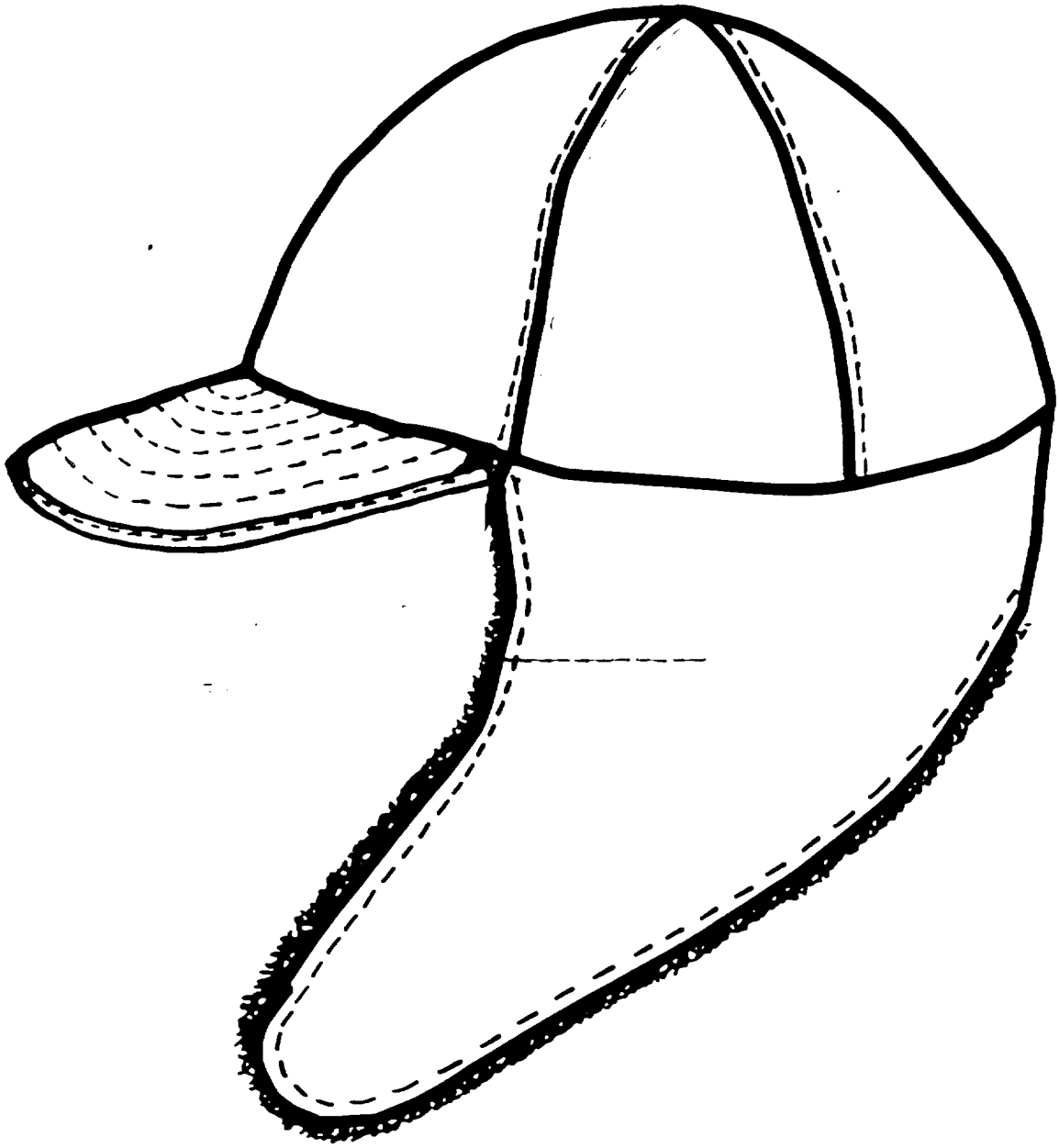


FIGURE 1:CAP

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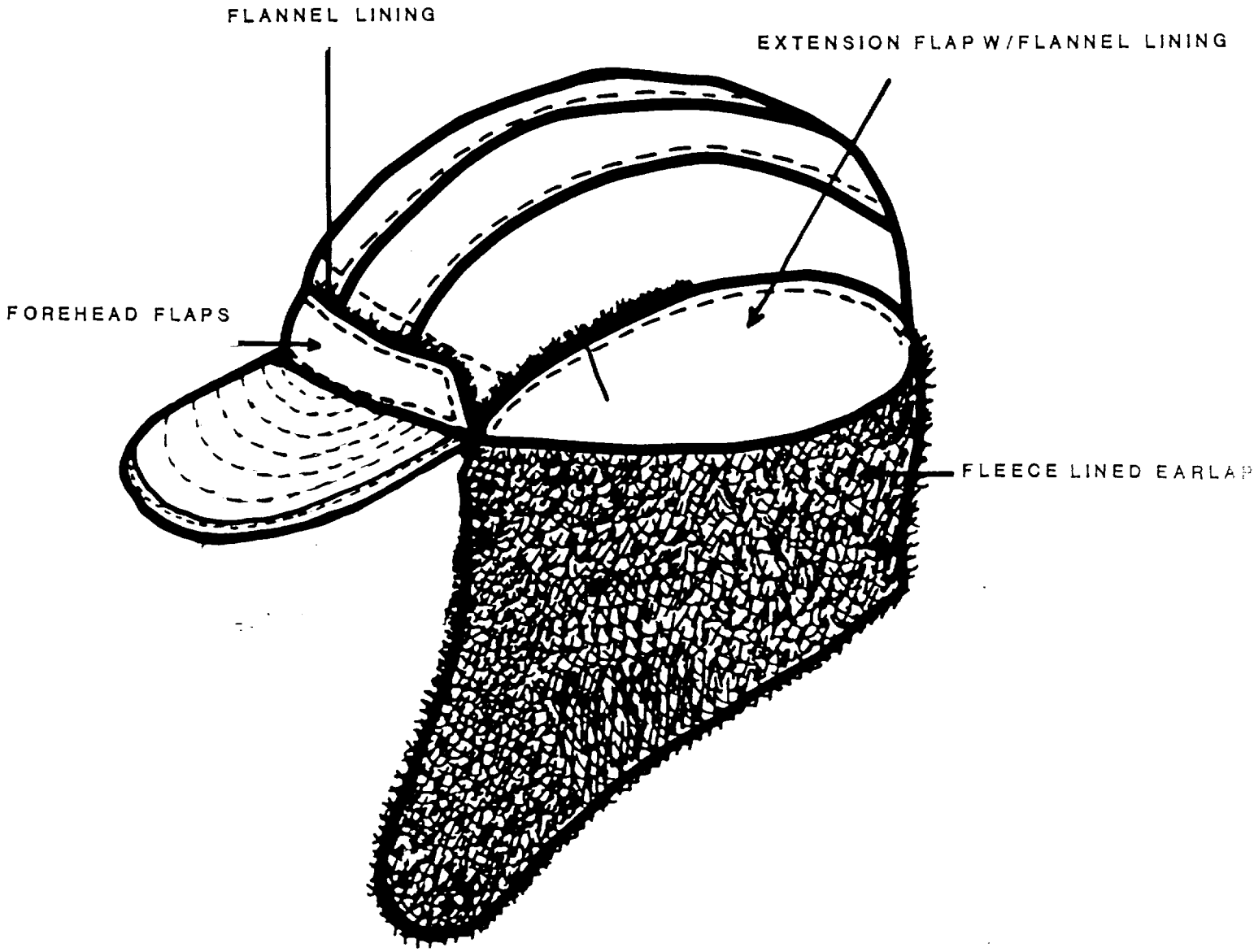


FIGURE 2: INSIDE OF CAP , EXTREME COLD WEATHER

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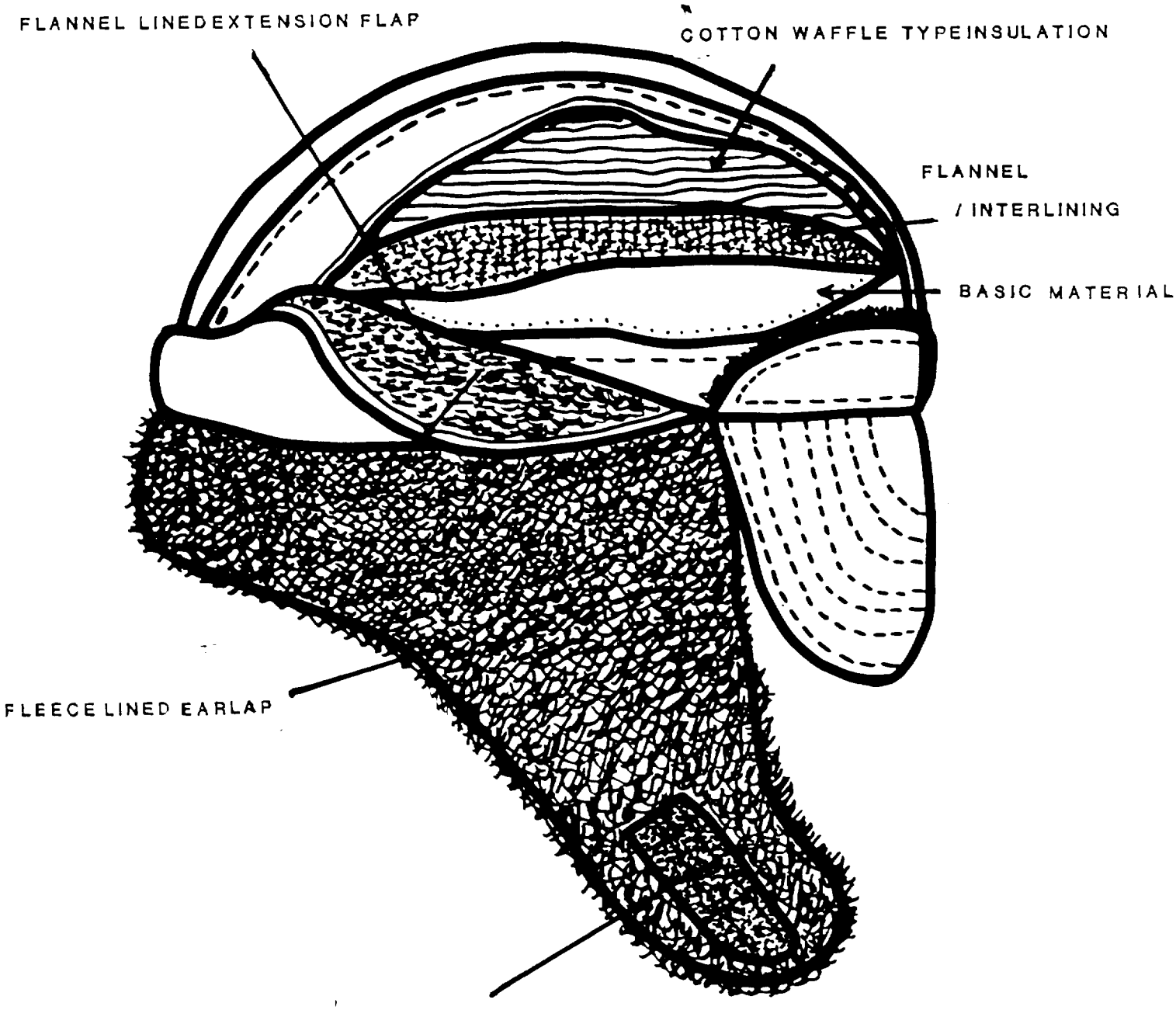
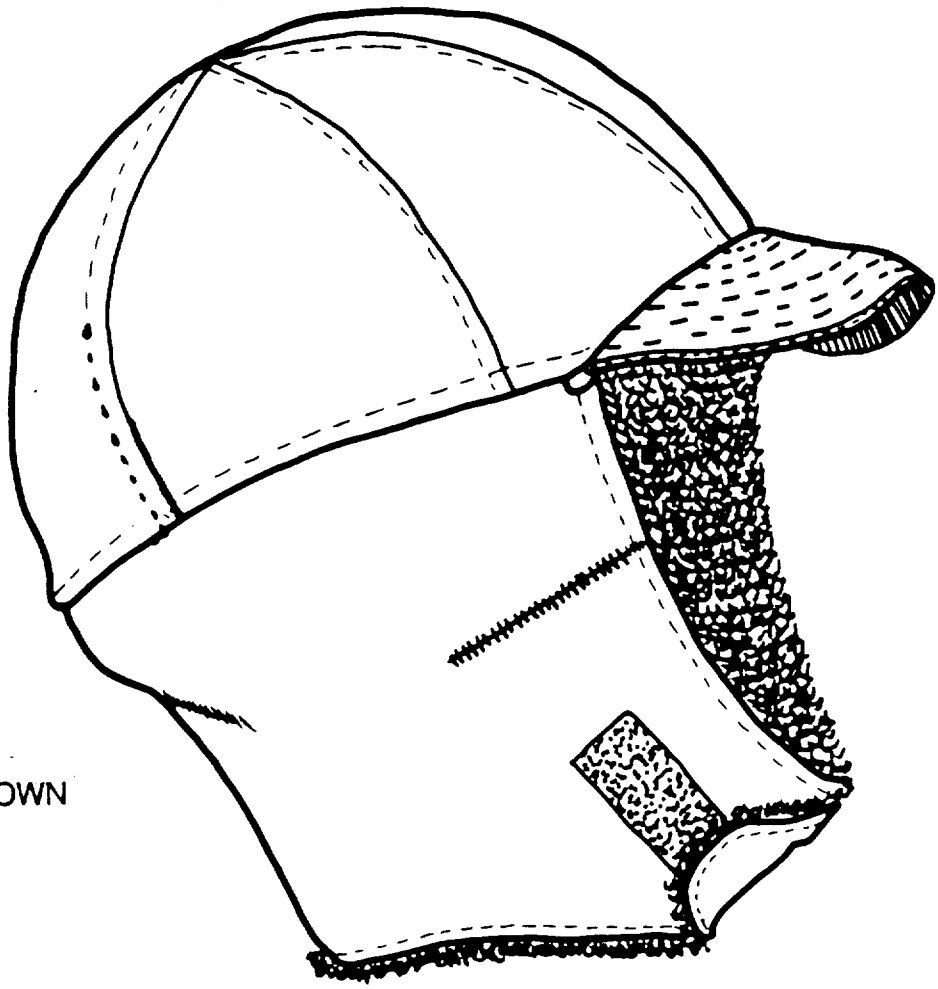
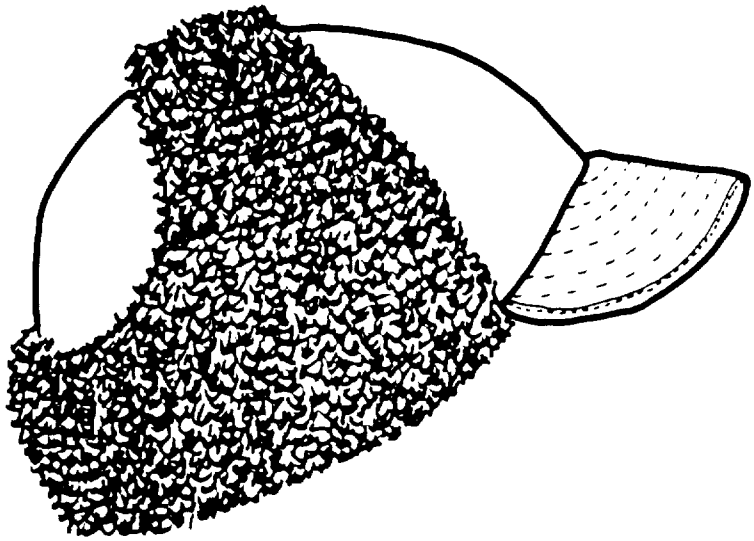


FIGURE 3:

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EARLAP DOWN

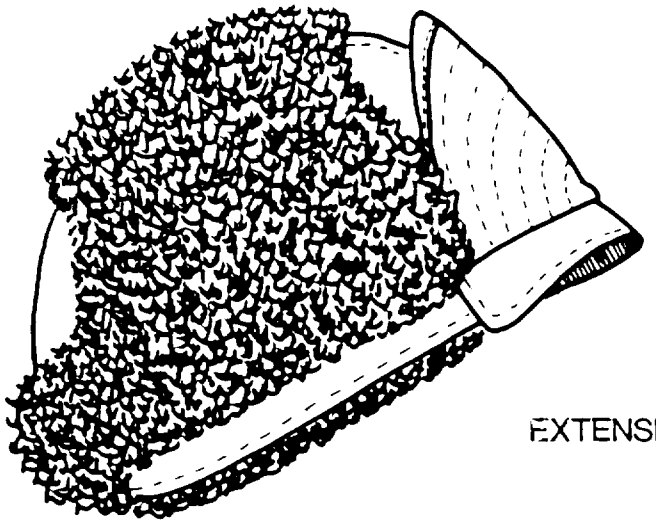


EARLAP UP

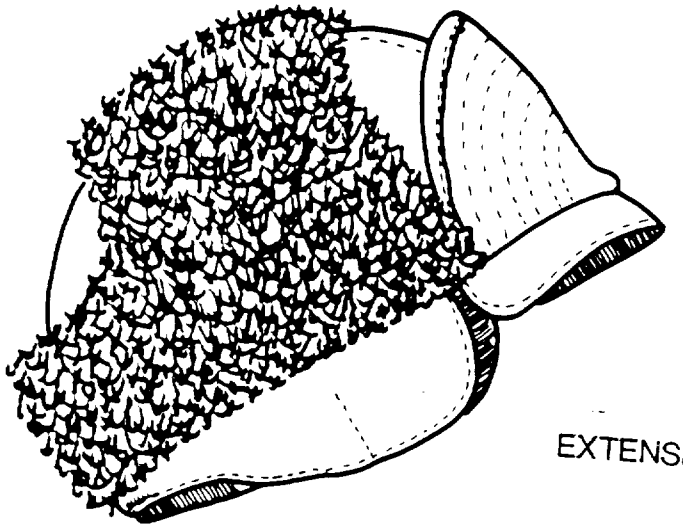
FIGURE 4

OUTSIDE VIEW

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EXTENSION FLAP UP



EXTENSION FLAP DOWN

FIGURE 5

OUTSIDE VIEW

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NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29430(MC)	CAP, EXTREME COLD WEATHER: WITH EXTENSION FLAP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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